

Problems of astronomy

X26 (2026) — English translation

Part A. Impact of the Earth and an asteroid (3.5 points)

This part discusses the potential impact of the Earth and an asteroid, and the possibility of preventing such an impact. In this part you may need the following data: Distance from the Earth to the Sun $r_a = 1.5 \cdot 10^{11} \text{ m}$. Consider the Earth's orbit to be circular. The radius of the Earth is $R_E = 6.4 \cdot 10^6 \text{ m}$. Earth mass $M_E = 6.0 \cdot 10^{24} \text{ kg}$. The mass of the Sun is $M_S = 2.0 \cdot 10^{30} \text{ kg}$. Gravitational constant $G = 6.67 \cdot 10^{-11} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2}$. Asteroid parameters: Diameter $D = 250 \text{ m}$. The density of the asteroid is $\rho = 2700 \text{ kg/m}^3$. The asteroid moves around the Sun in an elliptical orbit in the plane of the Earth's orbit. Eccentricity $e = 0.4$, the minimum distance to the Sun is equal to the radius of the Earth's orbit. The direction of rotation around the Sun coincides with the direction of rotation of the Earth.

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Asteroid parameters:

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| A1 | Find the orbital speed v_0 of the Earth's motion around the Sun, assuming the orbit is circular (formula and numerical value). Express your answer in terms of G , M_S , r_a . | 0.30 |
| A2 | Find the speed of the asteroid v_1 when it flies at its minimum distance from the Sun. Express your answer in terms of v_0 , e and give a numerical value. | 0.70 |
| A3 | It is known that at some point the asteroid will collide with the Earth. The collision is central. Find the energy released as a result of the collision (formula and numerical value). Express your answer in terms of the mass of the asteroid m_A , G , v_0 , e , R_E , M_E . | 0.60 |

Let's look at how to avoid such a collision. To do this, a bomb is delivered to the asteroid, as a result of which the explosion imparts an additional impulse Δp to the asteroid.

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| A4 | Let us assume that the asteroid is already at a sufficiently small distance from the Earth $s \ll r_a$, so that the influence of the Sun and other celestial bodies on its motion can be neglected. However, it is still $s \gg R_E$. Its speed in the Earth's reference frame (at a large distance to it) is v_r and is directed towards the center of the Earth. What is the minimum impulse $\Delta p_{\min}$ that must be imparted to it so that it does not collide with the Earth? Express your answer in terms of s , G , M_E , R_E , v_r and the mass of the asteroid m_A . For safety, the minimum distance to the center of the Earth must be at least $2R_E$. Consider that the transmitted momentum is much less than the momentum of the asteroid. | 1.40 |
| A5 | What is the minimum distance $s_{\min}$ at which the asteroid can still be deflected if the momentum change $\Delta p = 2 \cdot 10^{12} \text{ kg} \cdot \text{m/s}$ created by the bomb? Express your answer in terms of Δp , G , m_A , v_r , M_E , R_E and give the numerical value. | 0.50 |

Part B. Bertrand's Theorem (6.5 points)

In this part, we consider the motion of a particle of mass m in a central field, the potential energy of the particle in which is $U(r)$. It is well known that in the gravitational field of a point mass $U = -A/r$ bounded trajectories are ellipses and therefore closed. With an arbitrary dependence of potential energy on distance, it is also possible to find closed trajectories, for example circles. However, with a slight change in the initial conditions, the trajectories open up and fill a certain area of space. It turns out that it can be shown that the only potential decreasing at infinity for which all bounded trajectories are closed is the gravitational potential. This statement is called Bertrand's theorem. In all points of this part of the problem, the polar coordinate system r, θ is used. The mass of the particle m is assumed to be known. By oscillation frequency we mean cyclic frequency throughout.

Consider an arbitrary potential energy $U(r)$.

B1 Find the angular velocity ω_0 of the particle in a circular orbit. Express your answer in terms of m , the orbital radius R , $U(r)$ and its derivatives. **0.30**

B2 Next we will consider deviations of orbits from circular ones. Express the angular momentum of the particle L in terms of $r, \dot{\theta}$ and m . **0.20**

B3 Express the energy of the system E in terms of $L, m, r, \dot{r}$ and $U(r)$. **0.20**

B4 Let us introduce the effective potential energy $U_{\text{eff}}(r)$ such that the particle energy has the form: **0.10**

$$E = U_{\text{eff}}(r) + f(\dot{r}).$$

Запишите выражение для $U_{\text{eff}}(r)$.

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B5 Express the second derivative of the modulus of the radius vector with respect to time $\ddot{r}$ in terms of $m, U_{\text{eff}}(r)$ and its derivatives. Show that when moving in a circle, the effective potential energy U_{eff} reaches an extremum. **0.40**

Let us consider potential energy of the form $U(r) = -\frac{A}{r^\alpha}$, where $\alpha > 0$.

Let the particle move in such a potential field in a circular orbit of radius R . Let us consider a small deviation δr of the particle from the circle in the radial direction, so that $r = R + \delta r$. Consider the value of the angular momentum L of the particle to be the same as when it moves along a circle of radius R . A circular trajectory is stable if a sufficiently small deviation from it does not increase indefinitely with time.

B6 At what values of α is a circular orbit stable? Explore edge cases as well. Construct qualitative plots of the effective potential energy $U_{\text{eff}}(r)$ for various stable, unstable and boundary cases. Each graph must correctly reflect the qualitative behavior of the function (increase/decrease/convexity); calculation of the coordinates of characteristic points is not required. **2.00**

B7 Let us return to the case of arbitrary potential energy $U(r)$. Let the circular trajectory be stable. Similar to the previous point, consider a small deviation from the circular trajectory $r = R + \delta r$. Determine the frequency ω_r of the radial vibrations of the particle. Express your answer in terms of m , R , $U(r)$ and its derivatives. **0.80**

In order for a trajectory close to a circle to be closed, it is necessary that the ratio of the oscillation frequency to the angular velocity when moving in a circular orbit be a rational number: $\frac{\omega_r}{\omega_0} = q$ is a rational number. When the particle makes one oscillation in the radial direction (that is, r increases from the minimum distance to the maximum and decreases back to the minimum), the time $\Delta t = 2\pi/\omega_r$ will pass, and the change in the polar angle will be equal to $\omega_0 \Delta t = 2\pi/q$. Then, after an integer number of revolutions, the particle will perform an integer number of oscillations in the radial direction, and the trajectory will close.

B8 Using the above relationship between frequencies, obtain the differential equation for $U(r)$. **1.00**
The equation may also contain r and q . Show that this equation is satisfied by solutions of the form $U(r) = -\frac{A}{r^\alpha}$. We will only consider solutions with $\alpha > 0$. Derive an expression for the exponent α in terms of q .

The previous relation was obtained in the limit of infinitesimal deviations from a circular orbit. To determine the possible values of the exponent α , it is necessary to consider significant deviations of the trajectory from the circular one. The ratio of the frequency of radial vibrations to the angular velocity of rotation around the circle is a continuous function of E , L and other parameters of the system. On the other hand, for the trajectory to be closed, it must always remain a rational number. Therefore, this ratio, equal to q , is constant.

B9 Determine the change in the polar angle $\Delta\theta$ when moving from the trajectory point with the minimum value $r = r_{\min}$ to the nearest point with the maximum value $r = r_{\max}$. Express your answer as an integral over dr from the minimum distance $r_{\min}$ to the maximum distance $r_{\max}$. Your answer may also contain r , E , L , $U(r)$, m . Clue. Express $\dot{r}$ from the law of conservation of energy and $\dot{\theta}$ from the law of conservation of angular momentum, from here you can find the derivative $d\theta/dr$. **0.60**

Clue. Express $\dot{r}$ from the law of conservation of energy and $\dot{\theta}$ from the law of conservation of angular momentum, from here you can find the derivative $d\theta/dr$.

B10 Let us consider a convenient limiting case of motion with infinitesimal energy for the potential energy $U = -\frac{A}{r^\alpha}$. In this case, we can put $E = 0$, $r_{\max} = \infty$ in the integral and go to the variable $u = r_{\min}/r$. In this limit, express $\Delta\theta$ in terms of α . You will need the value of the integral: **0.60**

$$\int_0^1 \frac{du}{\sqrt{u^\alpha - u^2}} = \frac{\pi}{2 - \alpha}.$$

B11 As stated earlier, the ratio q will remain constant. Therefore, the angle change found above **0.30** must be equal to half the angle change during radial oscillation: $\Delta\theta = \frac{\pi}{q}$, where the movement from minimum to maximum is considered, and not the full oscillation. Using this fact, obtain an equation for α and find all its solutions.

Source: <https://pho.rs/p/4812>